

Lecture ~~2~~³: Finish Linear Regression, Regularized Regression

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Machine Learning I: 46-926

November 4, 2020

Announcements

- ▶ Current office hour schedule:
 - ▶ Monday, 10:20am: Me
 - ▶ Tuesday, 5pm: Vaibhav
 - ▶ Tuesday, 8pm: Wanshan
 - ▶ Thursday, TBD: Me
- 

How does this look? Feedback welcome.

- ▶ First homework has now been turned in! Let's talk about it!
- ▶ Homework 2 will come out by tomorrow morning and is due **Wednesday before class**.
- ▶ First quiz! Out Friday, due Sunday.
 - ▶ Simple, short questions.
 - ▶ Timed.
 - ▶ Your own work.

Today: Finishing basics, Linear Regression and Regularization

- ▶ Discuss the homework
- ▶ Last comments on linear regression
- ▶ Extending linear regression to work in high dimensions
 - ▶ Ridge Regression
 - ▶ Lasso

Review: Supervised learning

Observe $(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{iid}{\sim} \mathcal{P}$. Estimate $\hat{\mu}(x)$ so that $\mathbb{E}\mathcal{L}(Y, \hat{\mu}(X))$ is small for new $(X, Y) \sim \mathcal{P}$.

If we are minimizing expected squared prediction error:

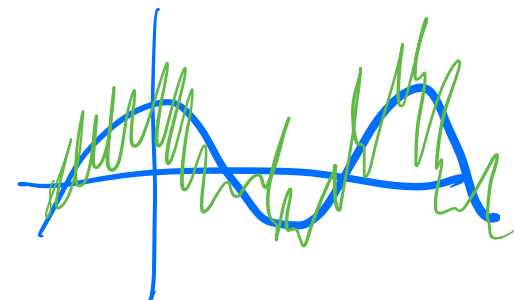
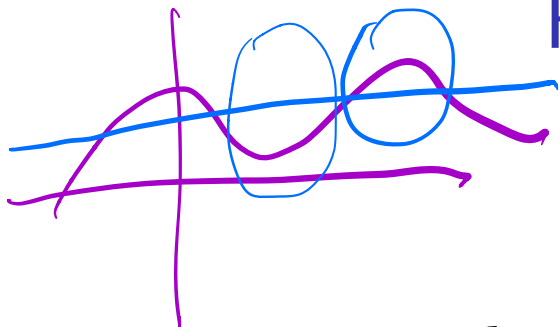
- If we actually knew $\mathcal{P}$, the best we could do is

$$\mu(x) = \mathbb{E}(Y|X = x)$$

- If we are estimating $\hat{\mu}$, then

$$\mathbb{E}(\mathcal{L}(Y, \hat{\mu}(X)) \mid X = x) = \sigma_x^2 + \text{bias}(\hat{\mu}(x))^2 + \text{var}(\hat{\mu}(x))$$

Review: Bias and variance



$$\text{bias}(\hat{\mu}(x)) = \mathbb{E}[\underbrace{\hat{\mu}(X)|X=x}] - \mu(x)$$

Bias: Error in the expectation of our estimator. Does not depend on the randomness of our particular training data realization, just the flexibility of our function. Making $\hat{\mu}$ more flexible usually decreases bias.

$$\text{var}(\hat{\mu}(x)) = \mathbb{E}\left[\underbrace{(\hat{\mu}(X) - \mathbb{E}[\hat{\mu}(X)|X=x])^2}_{\text{variance}} \mid X=x\right]$$

Variance: Error from the variance of our estimator around *its* mean. Does not depend on the true function μ ! Tends to increase with the flexibility of $\hat{\mu}$, since a change in data has more effect.

Review: Linear regression

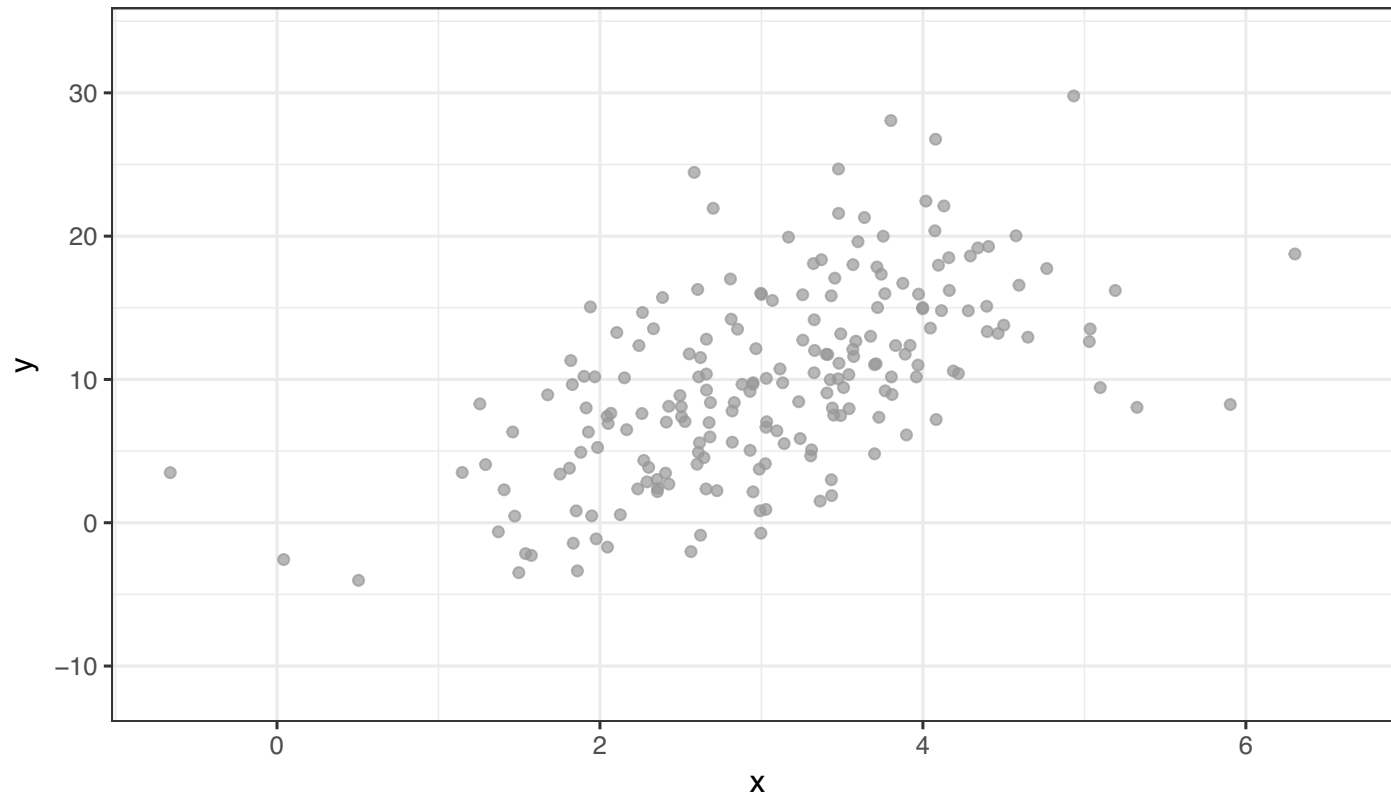
We can think of linear regression as one way to approximate $\mathbb{E}(Y|X = x)$ across x values, imagining:

$$\mathbb{E}(Y|X = x) \approx x^T \beta$$

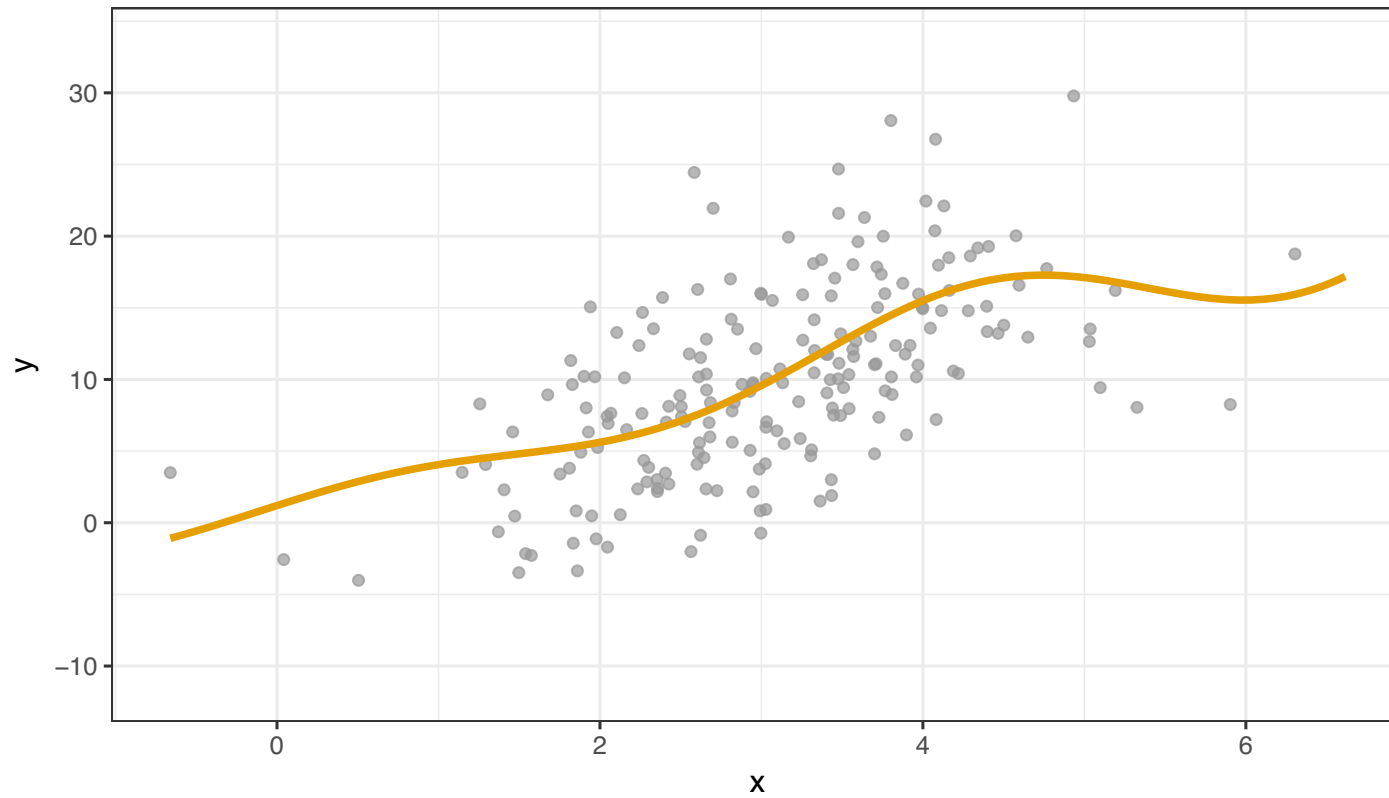
This can work well if the linear model is not a bad approximation.

It has the advantage of being very simple, which can behave better than very flexible models in high-noise settings.

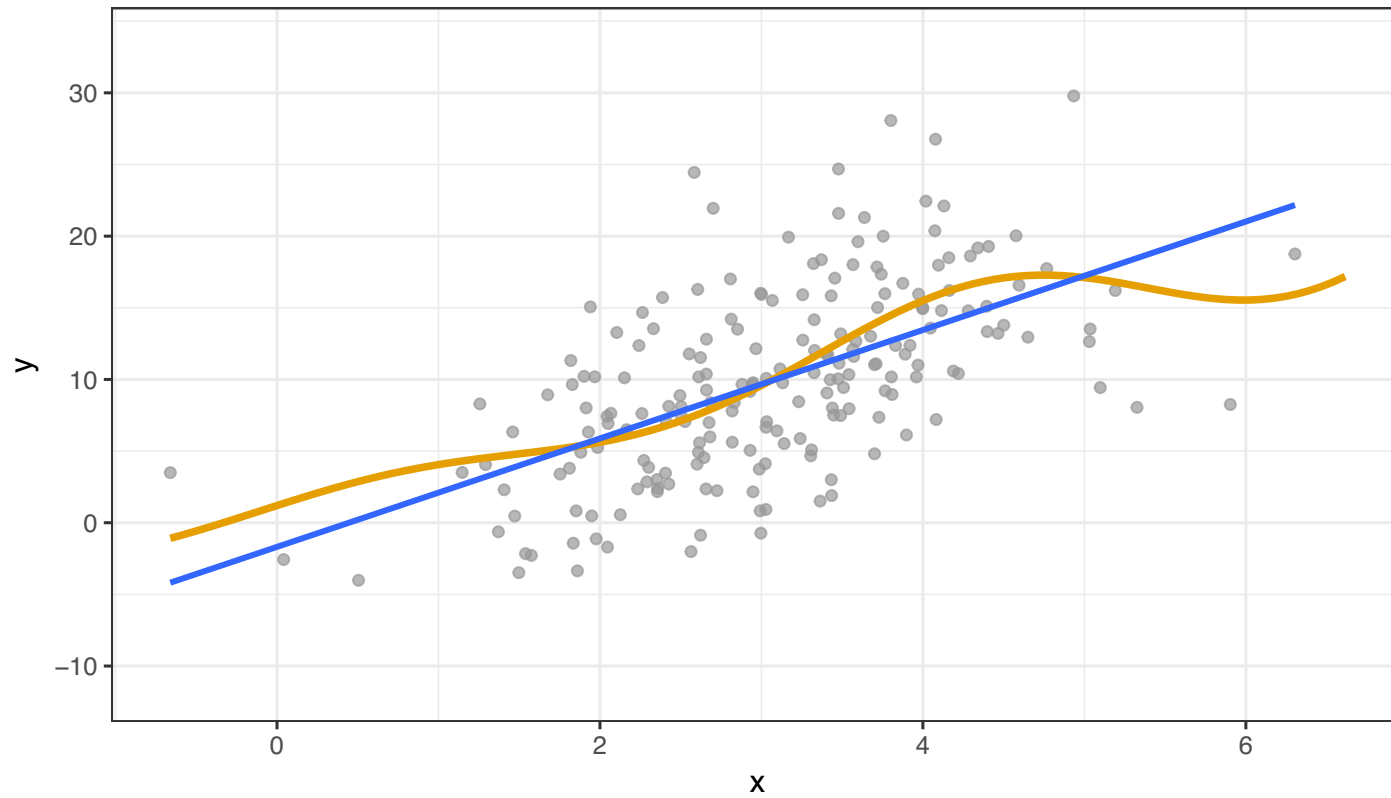
Regression function demo



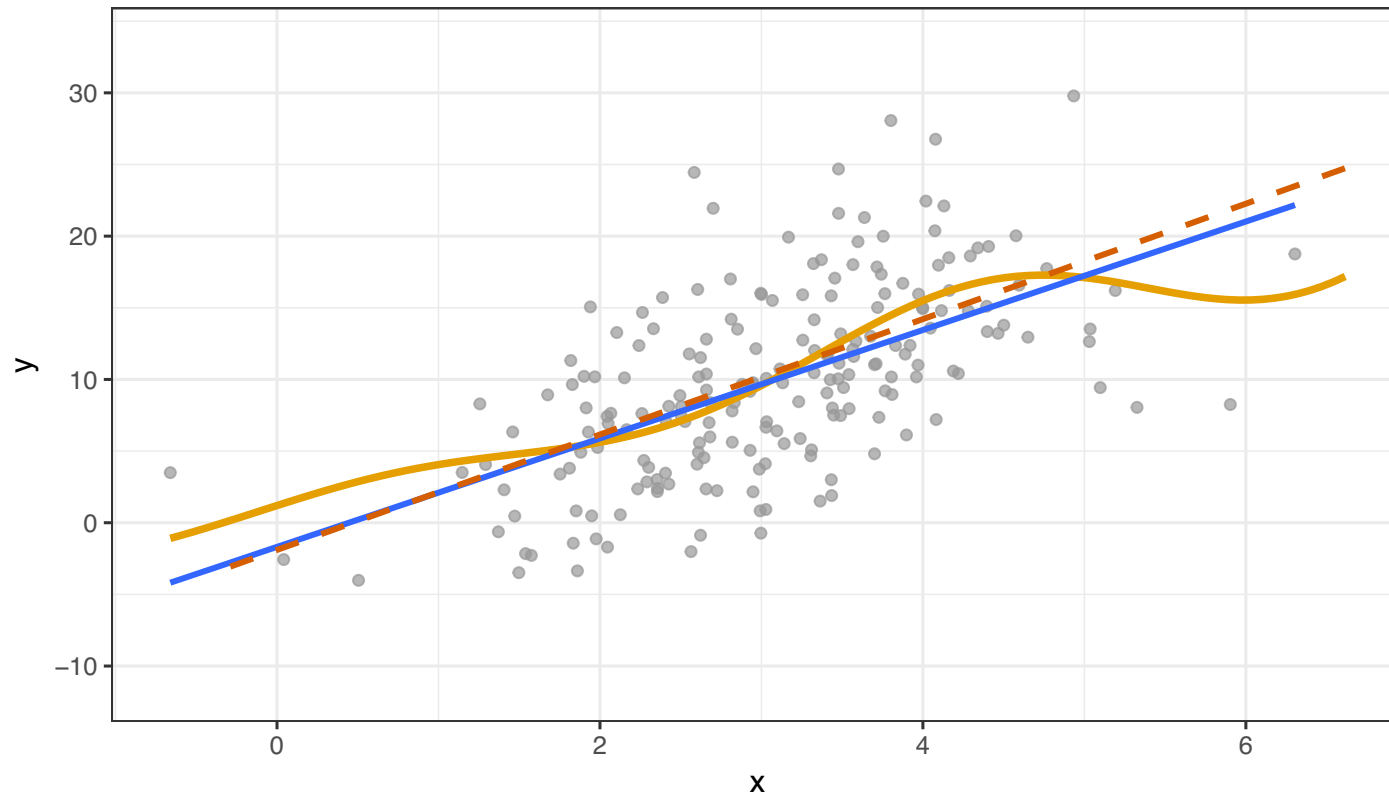
Regression function demo



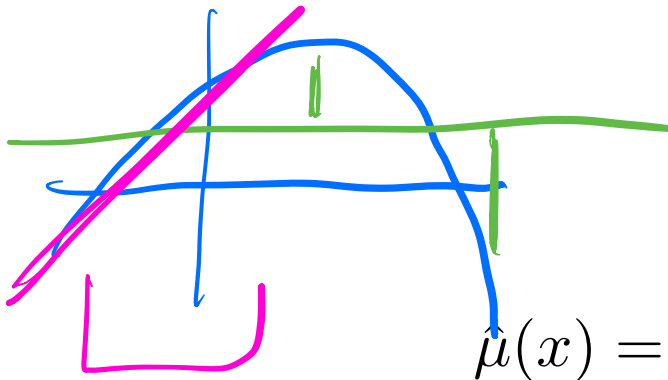
Regression function demo



Regression function demo



Linear regression



$$\begin{aligned}\hat{\mu}(x) &= \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p \\ &= x^T \hat{\beta}\end{aligned}$$

If the true $\mathbb{E}(Y|X)$ is very far from linear, this model will be highly *biased*, since even $\mathbb{E}\hat{\mu}(X)$ will never be able to match $\mathbb{E}(Y|X)$.

Even with infinite data, the best linear model would still be terrible. You learned (and we will discuss) much more flexible models to avoid this problem.

We will have other problems even when the linear model is correct!

Linear regression when the linear model is true!

Ok, so suppose that we use linear regression just to make predictions. Furthermore, suppose that the true conditional mean is linear! What will happen?

We know that least squares has nice properties when the linear model is true – e.g., it is unbiased and has minimum variance among unbiased, linear estimates (Gauss-Markov Theorem).

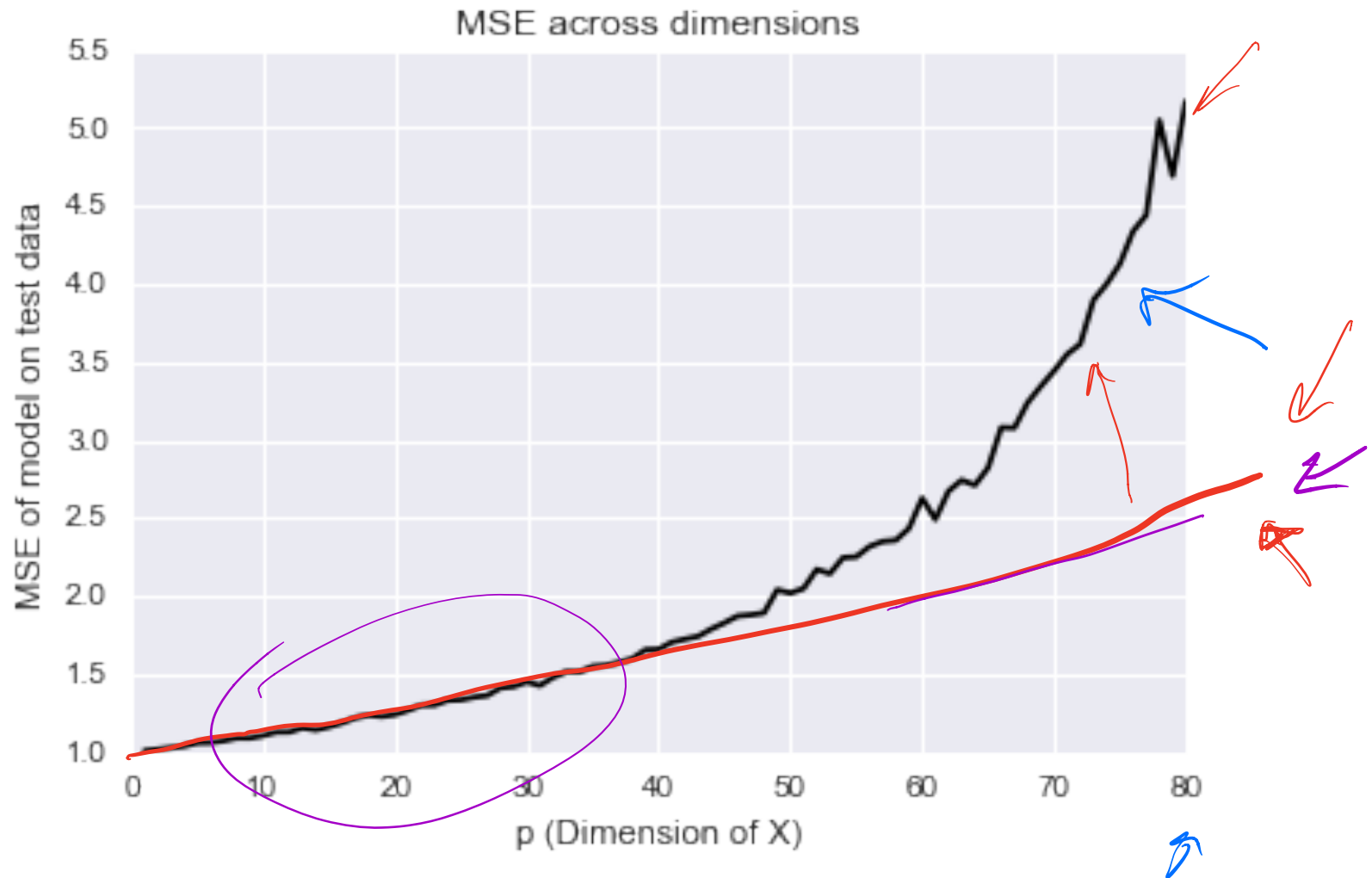
However...

Linear regression in high dimensions

What happens when we have a bunch of variables? (Big data!)

However, we noticed right at the end of class that as the number of variables, p , grows large, the variance of linear regression grows rapidly. This leads to poor predictions **even with the linear model is true!**

Linear regression in high dimensions



This is a simulation from a true linear model with independent Gaussian errors – the dream! (Also the plot from your homework.)

Linear regression in high dimensions

Suppose that the linear model is even correct, so that $\mathbb{E}(Y|X)$ is exactly linear. What happens when we try to estimate it?

As dimension increases, the prediction error increases (super-)linearly! When dimension is bigger than the number of observations, *Can't even fit OLS*

What is going on, and how can we fix it?

Two Pictures: n -space

$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \in \mathbb{R}^n \sim N(0, \sigma^2 I)$

$$\begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Views the columns $(x_{1j}, \dots, x_{nj})$ as vectors in $\mathbb{R}^n$, along with $(y_1, \dots, y_n)$ and $(\varepsilon_1, \dots, \varepsilon_n)$

$$Y = \sum_{j=1}^n X_j \beta_j$$

Easy to see:

$$\hat{y} = P_X[y] = P_X[\mu + \varepsilon_{\perp} + \varepsilon_{\parallel}] = \mu + 0 + \left[\varepsilon_{\parallel} \right] \frac{P}{n}$$

- How to fit our model
- What happens when new variables are added?
- How the error ε interacts with dimension?

Fixing linear regression in high dimensions

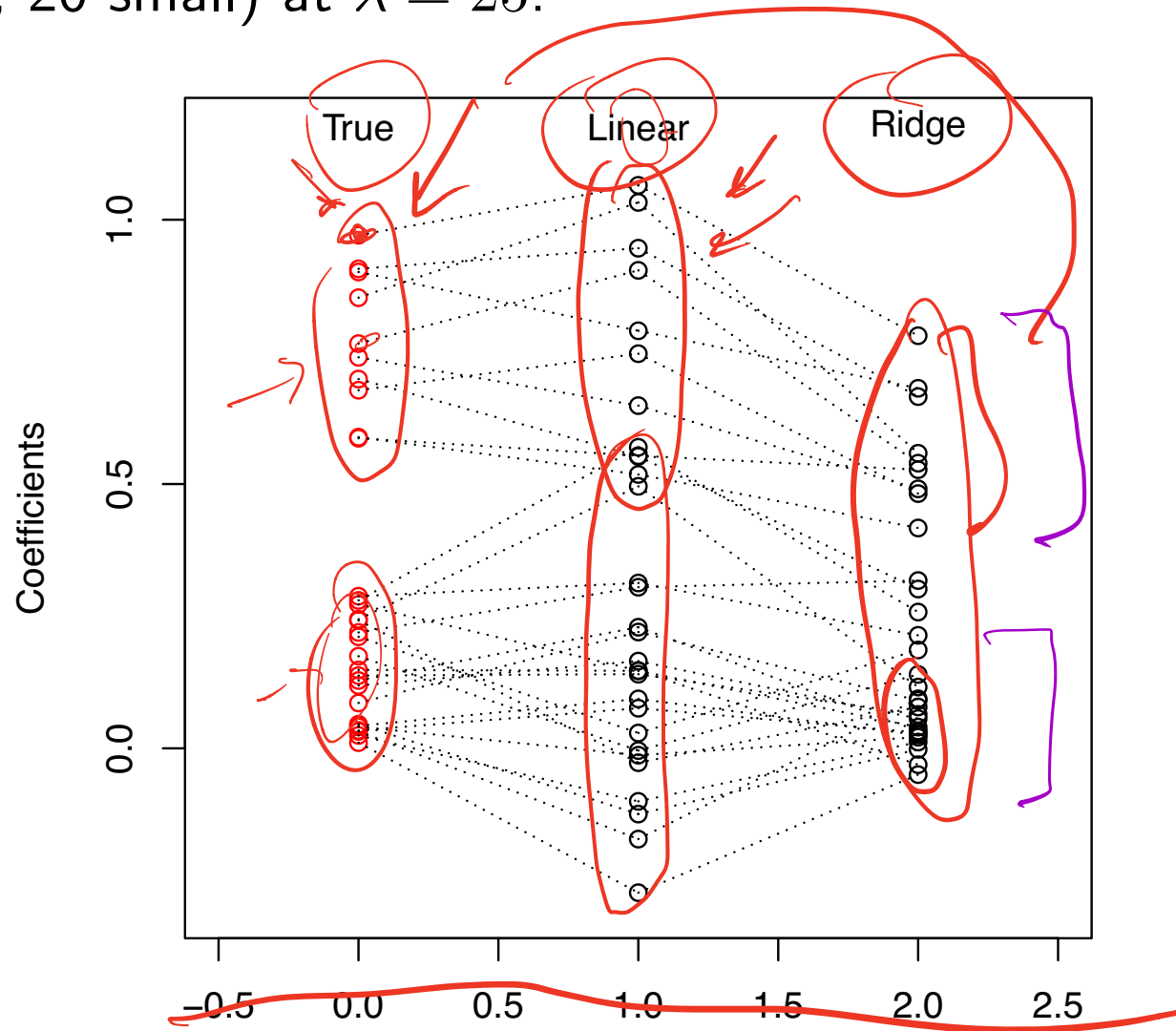
$$\hat{\mu}(x) = x\hat{\beta}$$

How do we reduce this error?

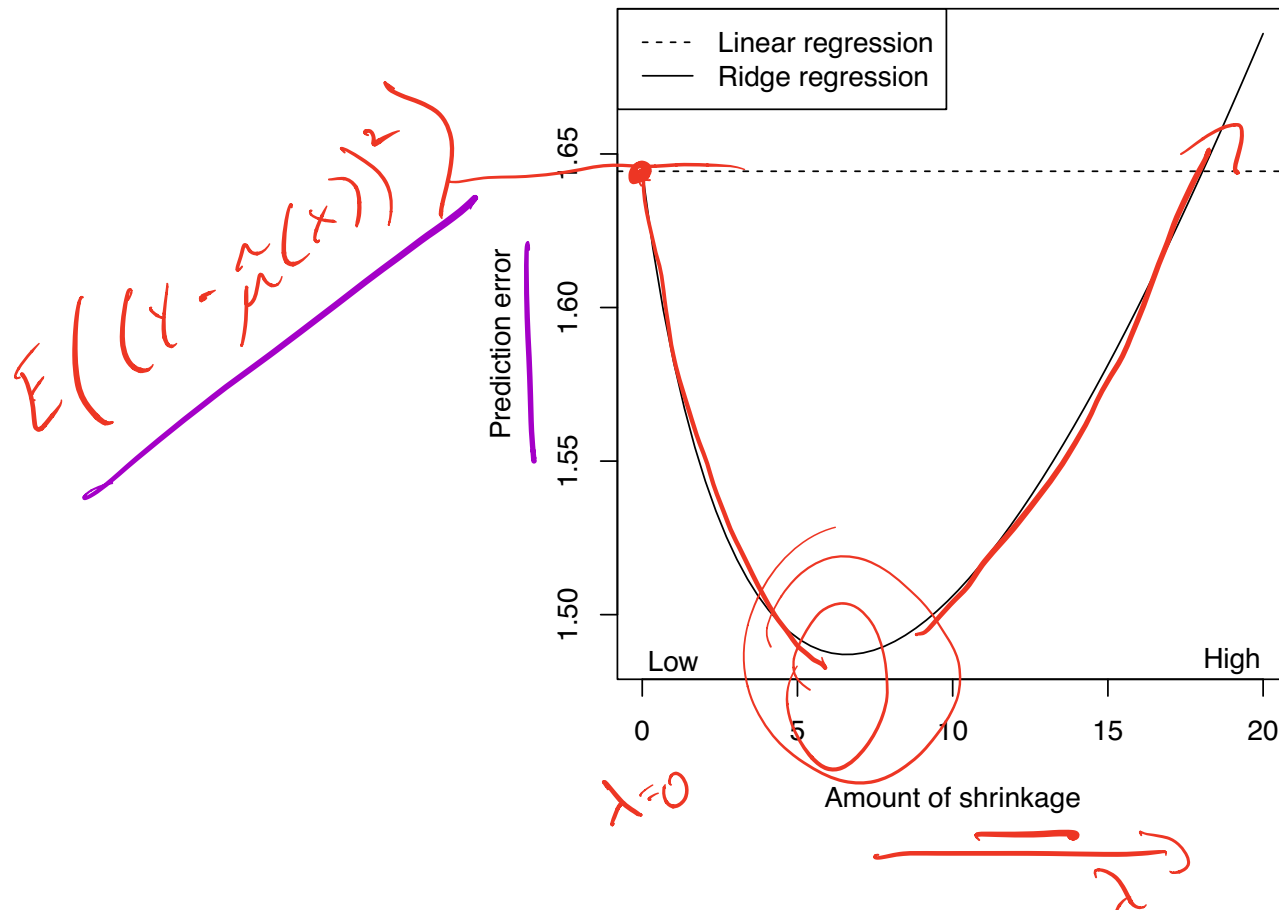
$$\mathbb{E} \left((Y - \hat{\mu}(X))^2 \right) = \sigma_x^2 + \text{Bias}(\hat{\mu}(X))^2 + \text{Var}(\hat{\mu}(X))$$

Example: visual representation of ridge coefficients

A visual representation of the **ridge regression** coefficients for the same example ($n = 50$, $p = 30$, and $\sigma^2 = 1$; 10 large true coefficients, 20 small) at $\lambda = 25$:



Does it work?



Linear regression:

Squared bias = 0

Variance ≈ 0.633

Pred. error $\approx 1 + 0.633$
 ≈ 1.633

Ridge regression, at its best:

Squared bias ≈ 0.077

Variance ≈ 0.403

Pred. error $\approx 1 + 0.077 + 0.403$
 ≈ 1.48

This is upsetting! And amazing!

James - Stein Estimator

$$y \in \mathbb{R}^n, \underbrace{y \sim N(\theta, \sigma^2 I)}_{\substack{\text{MLE} \\ \text{JS}}}$$

min

$$\frac{\|\hat{\theta} - \theta\|_2^2}{n}$$

$$\left\{ \sum_{i=1}^n (\hat{\theta}_i - \theta_i)^2 \right\}$$

$$\text{MLE} : \hat{\theta}_i = y_i \leftarrow$$

$$\text{JS} : \hat{\theta}_i = \left[\left(1 - \frac{(n-2)\sigma^2}{\|y\|^2} \right)_+ \right] y_i \leftarrow$$

$$(z)_+ = \max(0, z)$$

Ridge regression

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y \in \mathbb{R}^p$$

Ridge regression is like least squares but shrinks the estimated coefficients towards zero. Given a response vector $y \in \mathbb{R}^n$ and a predictor matrix $X \in \mathbb{R}^{n \times p}$, the ridge regression coefficients are defined as

$$\begin{aligned} \hat{\beta}^{\text{ridge}} &= \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \sum_{i=1}^n (y_i - \underbrace{x_i^T \beta}_{x_{i1}\beta_1 + x_{i2}\beta_2 + \dots + x_{ip}\beta_p})^2 + \lambda \sum_{j=1}^p \beta_j^2 \\ &= \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{Penalty}} \end{aligned}$$

$$\min_{z \in \mathbb{R}} (z-3)^2 = 0$$

$$\operatorname{argmin}_{z \in \mathbb{R}} (z-3)^2 = 3$$

Ridge regression

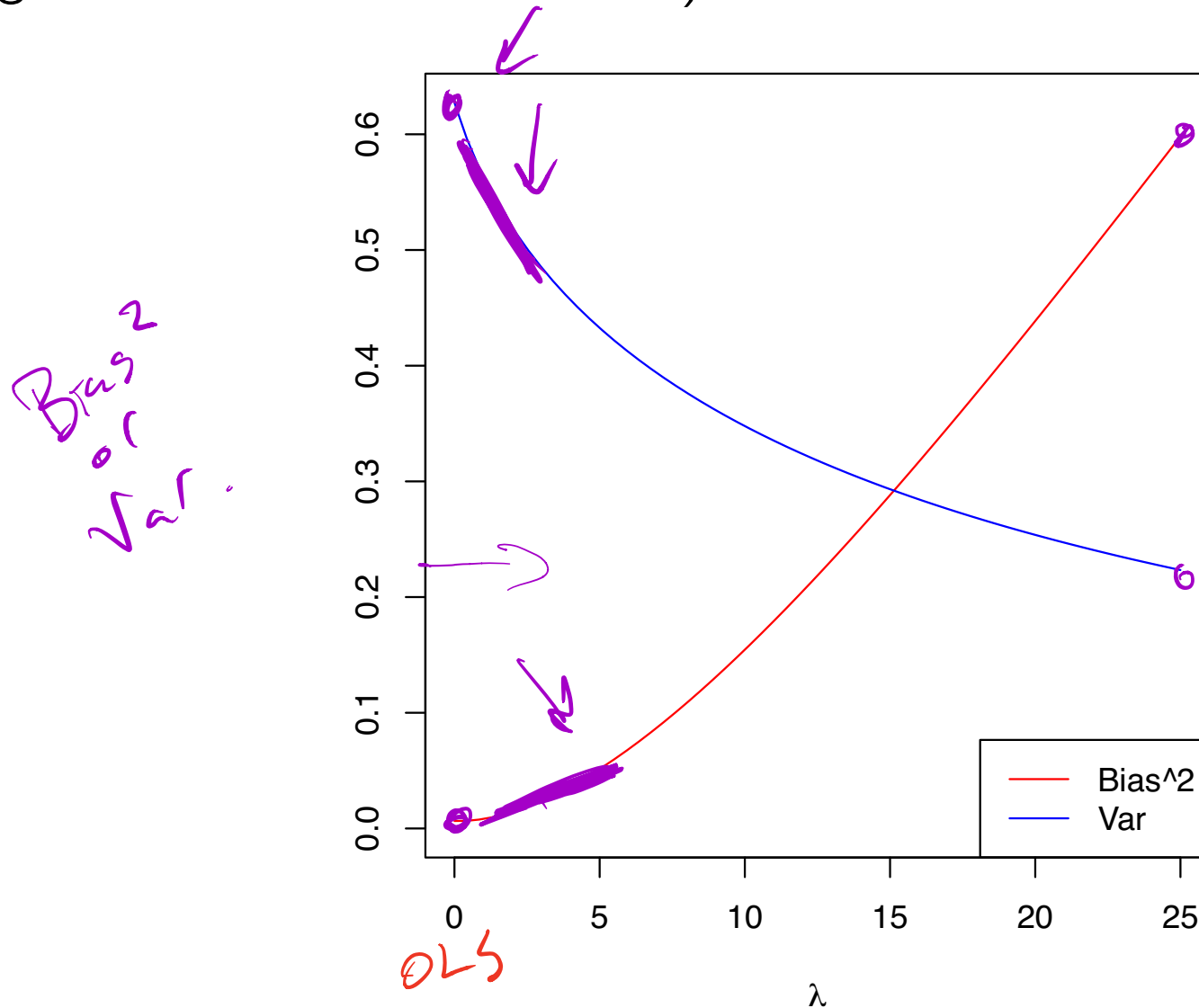
$$\begin{aligned}\hat{\beta}^{\text{ridge}} &= \operatorname{argmin}_{\beta \in \mathbb{R}^p} \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2 \\ &= \operatorname{argmin}_{\beta \in \mathbb{R}^p} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \underbrace{\lambda \|\beta\|_2^2}_{\text{Penalty}}\end{aligned}$$

Here $\lambda \geq 0$ is a **tuning parameter**, which controls the strength of the penalty term. Note that:

- ▶ When $\lambda = 0$, $\hat{\beta} = \hat{\beta}_{OLS}$ (it exists)
- ▶ When $\lambda \Rightarrow \infty$, $\hat{\beta} = 0$
- ▶ For λ in between, we are balancing two ideas: fitting a linear model of y on X , and shrinking the coefficients

Example: bias and variance of ridge regression

Bias and variance for our last example ($n = 50$, $p = 30$, $\sigma^2 = 1$; 10 large true coefficients, 20 small):



Important details: centering

When including an **intercept** term in the regression, we usually leave this coefficient **unpenalized**.

Otherwise we could add some constant amount c to the vector y , and this would not result in the same solution.

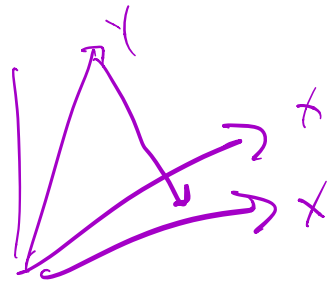
$$\sum \beta_i^2 \lambda_0$$

Hence ridge regression with intercept solves

$$\hat{\beta}_0, \hat{\beta}^{\text{ridge}} = \underset{\beta_0 \in \mathbb{R}, \beta \in \mathbb{R}^p}{\operatorname{argmin}} \quad \|y - \beta_0 \mathbf{1} - X\beta\|_2^2 + \lambda \|\beta\|_2^2$$

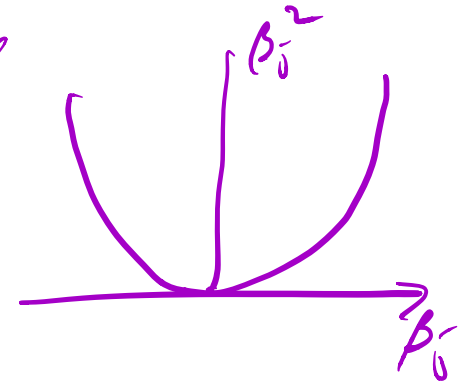
If we center the columns of X , then the intercept estimate ends up just being $\hat{\beta}_0 = \bar{y}$, so we usually just assume that y, X have been centered and don't include an intercept

Important details: scaling



$$\|Y - X\beta\|_2^2 + \lambda \sum \beta_j^2$$

Hand-drawn annotations: A bracket under the summation term $\sum \beta_j^2$ is labeled with β_j^2 . Arrows point from the β_j^2 in the equation to the β_j^2 in the bracket and to the β_j^2 in the adjacent graph.

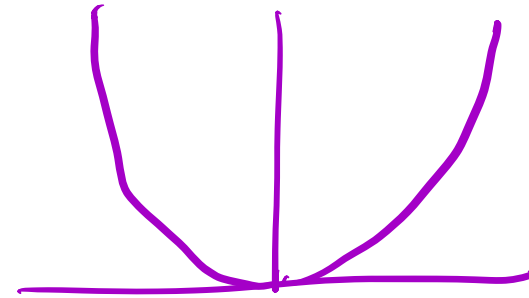


Also, the penalty term $\|\beta\|_2^2 = \sum_{j=1}^p \beta_j^2$ is unfair if the predictor variables are **not on the same scale**. (Why?)

Therefore, if we know that the variables are not measured in the same units, we typically **scale** the columns of X (to have sample variance 1), and then we perform ridge regression

We see that squishing the coefficients toward zero gives:

- ▶ A decrease in variance!
- ▶ A decrease in MSE!
- ▶ An increase in bias...

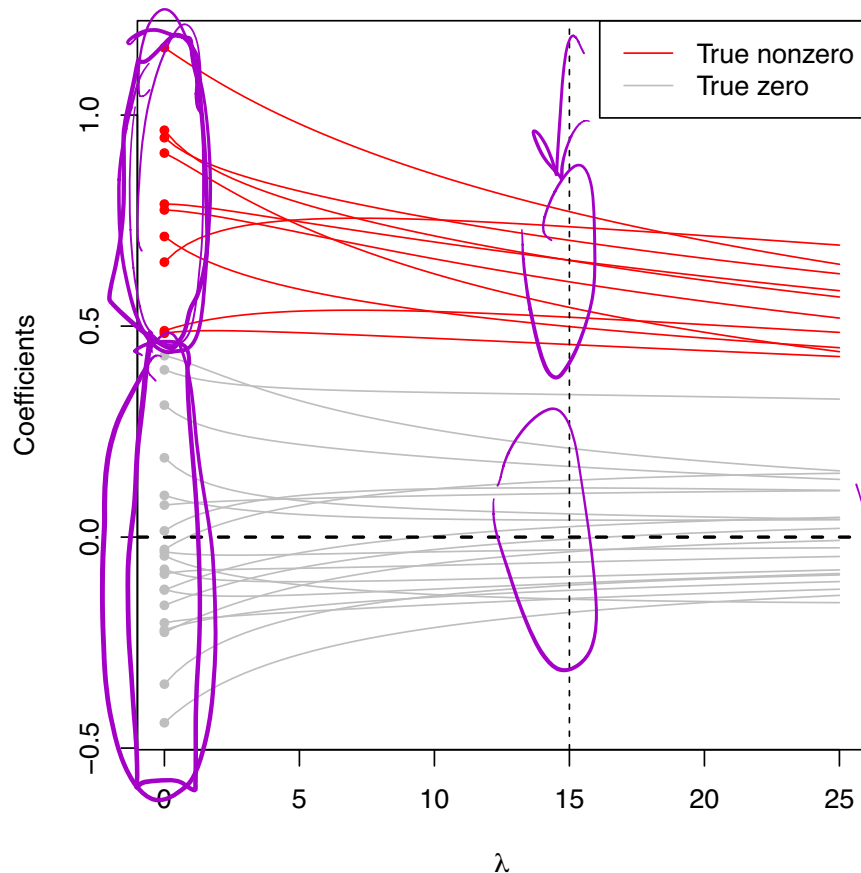


However, ridge regression:

- ▶ Seems quite harsh on large, important coefficients.
- ▶ Leaves us with a large linear model, which is not that interpretable when we have many variables.

Ridge regression coefficients

Ridge regression coefficients, plotted against λ :



The red paths correspond to the true nonzero coefficients; the gray paths correspond to true zeros. The vertical dashed line at $\lambda = 15$ marks the point above which ridge regression's MSE starts losing to that of linear regression

An important thing to notice is that the gray coefficient paths are not **exactly zero**; they are shrunken, but still nonzero